



Roundabout Dilemma Zone Data Mining and Forecasting with Trajectory Prediction and Graph Neural Networks

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Abstract

Traffic roundabouts, as complex and safety-critical road scenarios, present significant challenges for autonomous vehicles. In particular, predicting and managing dilemma zone (DZ) encounters at roundabout intersections remains a pivotal concern. This paper introduces an AI-driven system that leverages advanced trajectory forecasting to anticipate DZ events, specifically within traffic roundabouts. At the core of our framework is a modular, graph-structured recurrent architecture powered by graph neural networks (GNNs). By modeling agent interactions as a dynamic graph, our approach integrates heterogeneous data

sources - including semantic maps - while capturing agent dynamics with high fidelity. This GNN-based forecasting model enables accurate prediction of DZ events and supports safer, data-driven traffic management decisions for both autonomous and human-driven vehicles. We validate our system on a real-world dataset of roundabout intersections, where it achieves high precision with an exceptionally low false positive rate of 0.1. Our work highlights the potential of AI and graph-based deep learning methods for advancing roundabout safety, offering a robust step toward more reliable and intelligent intersection management in the era of autonomous transportation.

Introduction

Drivers often face dilemmas at traffic intersections, where they must decide whether to stop or proceed through the intersection [1, 2]. Yellow-light dilemma zone (DZ) is the area on a roadway where a driver may encounter such a dilemma. This dilemma can lead to dangerous situations, such as red-light running (RLR) or rear-ended crashes, particularly at high-volume or high-speed intersections [1]. Various studies have investigated the factors affecting drivers' decision-making processes, such as gap size, speed, and visibility [1, 3, 4].

Transportation engineers have proposed various solutions, such as advanced warning signs or pavement markings and intelligent transportation systems (ITS) using pressure, radar, or camera-based technologies to detect and monitor yellow-light DZs [5, 6, 7, 8]. However, these solutions are not applicable to roundabouts that do not have traffic lights. Instead, roundabouts use yield-controlled entries and continuous circular flows to improve traffic flow efficiency, reduce delays, and enhance safety. As a result, roundabouts can also create DZs where drivers may struggle to judge the safe gap size or time to enter the roundabout since vehicles inside the roundabouts create conflicts similar to "moving yellow lights",

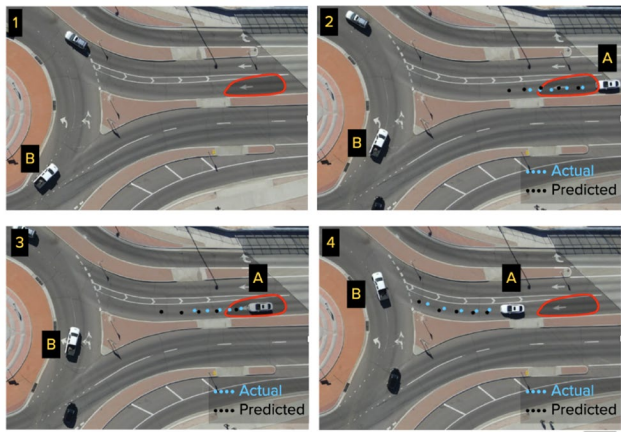
as shown in [Figure 1](#). In recent years, roundabout DZs have gained attention due to their unique characteristics and challenges [9], such as difficulties in obtaining accurate training data for driver behavior prediction [4, 10], changing factors affecting driver decision-making [1, 11], and tracking errors from roadside units [12].

To tackle the challenges associated with roundabout DZ, we propose a practical data mining approach to detect abnormal driving behaviors and filter out roundabout DZ events given the road geometry and vehicle trajectories. Our method compares the predicted trajectory with the actual trajectory, as shown in [Figure 2](#). Specifically, our trajectory prediction model incorporates multiple pieces of information to make its prediction, such as the driver's speed and acceleration, the position of other vehicles, and a semantic map of the environment. If there is a discrepancy between the predicted trajectory and the actual trajectory, it means that the driver has had to brake hard, accelerate rapidly, or make a sudden turn, which is indicative of the vehicle being in a DZ [11]. Further, using a graph representation of the vehicles, we extend the trajectory predictor to forecast DZ events on the fly from the perspective of an autonomous vehicle, which can help the vehicle to make a decision.

FIGURE 1 An illustration of a roundabout dilemma zone event. Here, a car in the yellow zone is insinuating a “moving yellow light”, creating a dilemma zone that influences the car in the red zone entering the roundabout. This scene is generated through simulation with real-world vehicle trajectories captured from drones.



FIGURE 2 An example of the driving behavior in a roundabout DZ: (1) Vehicle “B” inside the roundabout triggers a DZ as the area circled in red. (2) Vehicle “A” enters the DZ, and its trajectory is predicted. (3) Vehicle “A” rapidly decelerates, causing extensive deviation between the predicted trajectory and the actual trajectory. (4) Upon exit, vehicle “A” resumes, and its predicted trajectory agrees with the actual trajectory.



We also utilize accurate vehicle tracking through drone data [13], which provides a unique bird's eye view (BEV) perspective [14] that is not easily obtainable by other means. However, it is important to note that the use of drone data is not universal, and there are limitations to collecting data for extended periods. However, in the context of autonomous vehicles, the BEV representation can also be generated using data from the vehicles themselves [15]. Modern autonomous vehicles are typically equipped with advanced sensors, such as LiDAR, radar, and cameras, which can be used to create a comprehensive view of the vehicle's surroundings. By fusing data from these sensors, it is possible to generate a BEV perspective that is similar to the one provided by drones.

In summary, our contributions are as follows.

1. We introduce a data mining method to effectively identify roundabout DZ occurrences using vehicle trajectories. Our method leverages trajectory prediction and abnormal driving behavior patterns in the DZ.
2. We propose a dilemma forecasting method with a graph neural network that estimates the likelihood that a given vehicle encounters a DZ at every moment. It can also suggest proactive measures to ensure safety.
3. We demonstrate the effectiveness of our method through extensive experiments on a dataset of real-world roundabouts, providing insights into the practicality.

Background

Roundabouts are designed to improve safety and traffic efficiency, yet they introduce unique challenges for drivers approaching the yield line. One of these challenges is the *roundabout dilemma zone (DZ)*, a region where drivers must quickly decide whether to yield or proceed based on the speed and position of circulating vehicles. Unlike traditional signalized intersections, the roundabout DZ involves continuous traffic flow and right-of-way conflicts, making driver decisions more dynamic and complex. Understanding the mechanics of this dilemma and modeling the associated vehicle behavior are essential for improving safety and designing intelligent decision-support systems for roundabout entry.

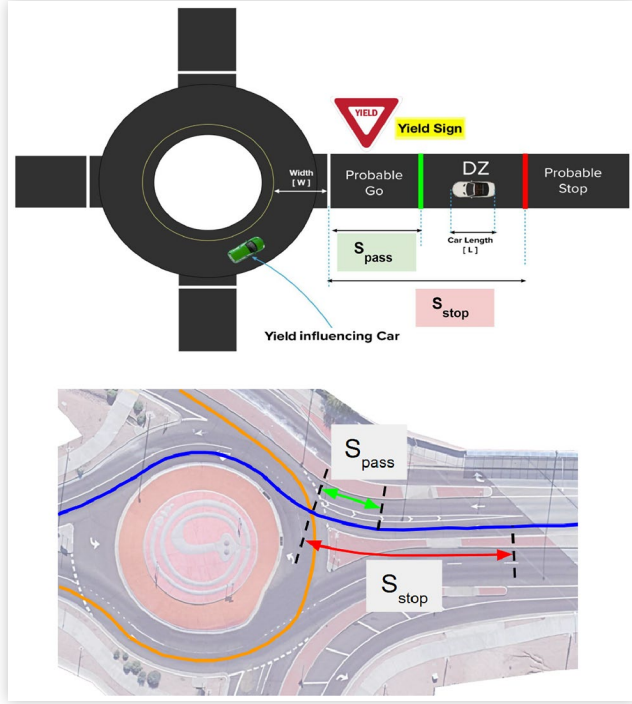
Roundabout Dilemma Zone

The roundabout DZ [2, 11, 16] is a brief period when a vehicle must decide whether to yield or pass when approaching a roundabout, based on the presence of other vehicles inside the roundabout. This can be a tricky decision for drivers, as it requires good judgment of the speed and distance of other vehicles. Improper judgments can result in traffic violations and, in some cases, even T-bone crashes [7]. Decision-making can be challenging, particularly with large traffic volumes or when traffic is moving at high speeds.

Figure 3 shows a vehicle in a dilemma zone. This mathematical model is adapted from the yellow-light dilemma zone [1, 2] by introducing a virtual yellow light generated by a yield-influencing vehicle inside the roundabout, following the standard yellow-light signal logic [17]. Specifically, if the driver's reaction time is given by δ_{react} and the vehicle is traveling at an initial velocity v_0 and accelerating at a_{acc} m/s, then the maximum distance at which the vehicle can pass safely is given by:

$$S_{pass} = W + L + \frac{1}{2} a_{acc} \delta_{react}^2 \quad (1)$$

FIGURE 3 Formulation of the roundabout DZ. The upper one shows an ideal general illustration with straight distance, while the lower one shows a realistic illustration with actual vehicle trajectories and distances along the curved trajectories.



Where W is the width of the road in the intersection, which accounts for the space needed for the car to maneuver within the roundabout without encroaching on adjacent lanes or objects, and L is the length of the vehicle subject to the dilemma, which accounts for the space required by the entire vehicle to pass through the roundabout. The generation of virtual yellow lights is detailed in later sections.

Similarly, if the vehicle has a safe braking deceleration of a_{dec} m/s, the minimum distance before it needs to stop is given by:

$$S_{stop} = v_0 \delta_{react} + \frac{v_0^2}{2a_{dec}} \quad (2)$$

We design S_{pass} as a conservative lower bound on the maximum yellow-light passing distance with a shift of the origin from the traffic light stop line to the roundabout trajectory conflict point [1, 2]. Because the virtual yellow light is dynamically generated, we do not have an exact yellow-light interval time. Instead, we estimate a lower bound by setting the interval time to zero. Additionally, the lane width and vehicle length are added to the distance, rather than subtracted, in order to filter out vehicles already entering the roundabout. On the other hand, S_{stop} is the same as the yellow-light minimum safe stopping distance. Moreover, the reaction time δ_{react} may be tuned to different values for passing and stopping.

A vehicle is said to be in a dilemma if it lies between the distances $S_{stop} - S_{pass}$ at a given virtual yellow-light

signal. The case when $S_{stop} > S_{pass}$ is a dilemma zone, whereas the case when $S_{stop} < S_{pass}$ is called an option zone. Generally, our formulation covers a larger DZ compared to traditional yellow light DZ. Calculating the roundabout DZ in our formulation requires the entire trajectory pair and vehicle-order information, which can only be achieved after the trajectories are recorded.

The behavior of vehicles in yellow-light DZs at intersections has been widely studied [10, 11, 18], and several factors were brought to light on the cause and effect. The three major factors to be considered are (a) approaching velocity, (b) acceleration after the event starts, and (c) the type of vehicle incurring the yield. In one study on yellow light DZs at signalized intersections [11], 26% of their drivers chose to stop, whereas 74% of their drivers were observed to run through the intersection. After the yellow signal started, the majority of vehicles elected to avoid the intersection, stopping only when there was a strong likelihood that they would commit a Red Light Running (RLR) violation. In another simulated study [18], it was observed that drivers stopping in a DZ decelerate rapidly, which could contribute to rear-end collisions.

While some studies have examined dilemmas at roundabouts, such as the one conducted by Najmi et al. [16], they do not fully extend the concept of normal DZs to roundabout DZs due to the significant presence of deceleration and priority given to circular traffic flow. The study also found that the dilemma zone in roundabouts was shorter in length and closer to the stop line. We extend the studies presented in existing works [10, 11, 18] to roundabouts by understanding the behavior of vehicles and the effect of the yield sign on them. We particularly solve this issue by generalizing a roundabout as an intersection with virtual signals and by predicting the trajectories of the vehicles present in the scene. This system can help drivers make informed decisions when approaching a roundabout, reducing the risk of accidents and improving traffic flow.

Trajectory Prediction

Social connections and spatial environment [19, 20] can affect how people and vehicles will move in the future. In the case of the DZ, the behavior of vehicles is characterized by vehicle dynamics. Works that consider dynamics [21, 22] use a graph-based approach to model the agent environment.

In particular, the Trajectron++ [21] models traffic scenes as a dynamic spatiotemporal graph whose nodes are agents (e.g., vehicles, pedestrians) with past state histories and whose edges represent time-varying interactions; the input consists of each agent's observed trajectory over a history window together with agent types, interaction features, and additional heterogeneous context such as map/lane geometry and agent-specific dynamics constraints. To encode this roundabout topology, the model processes rasterized semantic maps through a Convolutional Neural Network (CNN) to extract

local geometric features, which are then integrated into the agent's node representation to support generalization across different intersection layouts.

The output is a probabilistic, multi-modal distribution over future trajectories for each agent over a prediction horizon. Prediction is performed by encoding agent histories and interaction edges with recurrent networks to form a social context, sampling a latent intent variable within a conditional variational autoencoder (CVAE) to capture multiple plausible futures, and decoding this latent representation into dynamically feasible future motion distributions, with Trajectron++ extending the original framework by enforcing kinematic feasibility, better conditioning on map and agent-type information, and producing more realistic, physically consistent multi-agent trajectory forecasts. In this work, we employ a modified Trajectron++ to detect abnormal driving behaviors in the roundabout DZ.

Roundabout Dilemma Zone Data Mining

To effectively analyze driver behavior near roundabouts, it is crucial to identify when vehicles enter a dilemma zone and to detect the abnormal driving events that follow. This section presents a two-stage framework: first, determining the presence of a dilemma zone using vehicle dynamics and virtual signal logic, and second, detecting abnormal driving patterns that arise during these events through trajectory prediction. Specifically, the first step requires exhaustive trajectory pair analysis at known scenes, which is slow but accurate. The results of this first stage are leveraged to train a model for the second stage, which uses a trajectory predictor and a shallow neural network to determine abnormal driving behavior given the difference between the predicted trajectory and the recorded ground truth trajectory. This second stage only requires a tracklet or a segment of the trajectory of the subject vehicle, which is fast but less accurate. However, we can always apply the first stage again on the detection results to further check whether the detected abnormal events are roundabout DZ events or not. Together, these methods enable a systematic approach to uncovering and characterizing dilemma-zone occurrences in a real-world traffic dataset that can easily exceed hundreds of hours.

Dilemma Zone Determination

To unearth DZ data concealed within datasets of vehicle trajectories, we employ [equations \(1\) and \(2\)](#) designed to estimate the boundaries of the dilemma zone on the road surface. By plugging in the speed limits of the intersection, the safe acceleration, and the deceleration rate into this equation, we can calculate an approximate region where DZ events are likely to occur.

One pivotal aspect of our data mining strategy involves when to apply the calculation with the two

equations. Compared to yellow-light DZ at signalized intersections, we developed a procedure to determine when a vehicle inside the roundabout generates a “virtual yellow light” for vehicles approaching the roundabout, which is shown as Algorithm 1 in [Figure 4](#). It is based on driving safety metrics [\[23, 24\]](#), including time to conflict (TTC) and time to stop (TTS). There are two key observations as follows.

- **Identification of Virtual Signals:** A vehicle inside the roundabout requires vehicles entering the roundabout to yield, which is similar to a yellow light (when the vehicle inside is before the conflict point) or a red light (when the vehicle inside is at the conflict point). A yellow light indicates slowing down and proceeding with caution, while a red light indicates stopping and waiting due to trajectory conflict.
- **Assumption of True Dilemma:** Any vehicle exhibiting a yellow light signal within the vicinity of the calculated DZ is considered to be a true DZ event. Under such a DZ, the vehicle approaching the roundabout will show a different driving behavior compared to vehicles without the influence of DZ events. The proximity of the vehicle to the DZ boundary, within a distance denoted as D_t signifies their involvement in the event.

In Algorithm 1, there are two crucial hyperparameters. The first one is T_{max} , which represents the maximum acceptable time that the vehicle outside the roundabout

FIGURE 4 The virtual yellow light generation algorithm.

Algorithm 1 Virtual Yellow Light Generation

```

1: procedure CALCULATETRAFFICSIGN
2:    $Signal \leftarrow \text{Green}$ 
3:   for  $car_i$  approaching the roundabout do
4:      $pos_i \leftarrow \text{Position}(car_i)$ 
5:     for  $car_j$  in the roundabout do
6:        $pos_j \leftarrow \text{Position}(car_j)$ 
7:        $TTC \leftarrow \text{TimeToCollision}(car_i, car_j)$ 
8:        $TTS \leftarrow \text{TimeToStop}(car_i)$ 
9:        $SOC \leftarrow \text{SeparationOf}(car_i, car_j)$ 
10:       $t \leftarrow \text{Current time}$ 
11:       $T \leftarrow \text{Time of entry of } car_i \text{ into roundabout}$ 
12:       $T_{max} \leftarrow \text{Time limit for safe entry}$ 
13:       $D_t \leftarrow \text{Distance limit for influence}$ 
14:      if  $(TTC < T_{max}) \wedge (SOC < D_t)$  then
15:        if  $(TTC < TTS)$  then
16:           $Signal \leftarrow \text{Red}$ 
17:        return  $Signal$ 
18:      else if  $(TTC > TTS)$  then
19:         $Signal \leftarrow \text{Yellow}$ 
20:      end if
21:    end if
22:  end for
23:  return  $Signal$ 
24: end for
25: end procedure

```


has to reach and pass the conflict point. Given a specific vehicle outside the roundabout, if the time required for this vehicle to pass the conflict point exceeds T_{max} , it is considered that the vehicle inside the roundabout and the vehicle outside the roundabout have enough separation such that the interaction of these two vehicles incurs no dilemma. The second one is D_t , which represents the minimum separation distance between a vehicle pair that can potentially involve a dilemma. Given a vehicle inside a roundabout, it only influences those vehicles within D_t . Both hyperparameters help us quickly filter out irrelevant vehicle pairs.

We use this DZ determination method to obtain ground-truth DZ events to train our abnormal driving event detector, detailed in the next subsection. More details are explained in our technical report [28].

Abnormal Driving Event Detection

Predicting the trajectory of a vehicle on the road would serve as the ground for detecting dilemma events. Salzmann et. al [21] proposed a forecasting model that was capable of incorporating agent dynamics with semantic maps (heterogeneous data) in a multi-agent scenario. We propose Superpowered Trajectron++ (ST++), which is a modified version of Trajectron++ [21] trained with the features that have a significant impact [2, 11] on the dilemma zone to predict a vehicle's trajectory under non-dilemma conditions. Under a dilemma zone, the behavior of the driver changes drastically, which is abnormal.

To leverage this knowledge, a trajectory prediction model that produces a target probability $p(\mathbf{y}|\mathbf{x})$ where

$$\mathbf{x} = \mathbf{s}_{1,...,N(t)}^{t-H:t} \in \mathbb{R}^{(H+1) \times N(t) \times D}$$

are the current and previous D-dimensional states of the modeled agents are trained under non-dilemma conditions.

The model outputs the probability distribution of the output trajectory

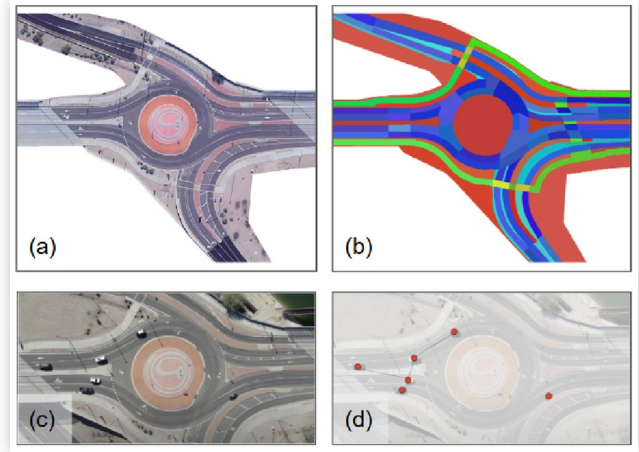
$$p(\mathbf{y}|\mathbf{x}) = \sum_{z \in Z} p_{\psi}(\mathbf{y}|\mathbf{x}, z) p_{\theta}(z|\mathbf{x})$$

where $z \in Z$ is a discrete Categorical latent variable that encodes high-level latent behavior, ψ and θ are deep neural network weights that parameterize their respective distributions. The most likely trajectory $\mathbf{y}_{pred}$ can be calculated from the above distribution:

$$\begin{aligned} z_{mode} &= \operatorname{argmax}_z p_{\theta}(z|\mathbf{x}), \\ \mathbf{y}_{pred} &= \operatorname{argmax}_{\mathbf{y}} p_{\psi}(\mathbf{y}|\mathbf{x}, z_{mode}) \end{aligned} \quad (3)$$

The Superpowered Trajectron++ makes predictions with two pieces of information: a static semantic map and a dynamic vehicle relation graph, shown in Figure 5. Since the trajectory prediction is done for a subject

FIGURE 5 The map and vehicle relation graph: (a) the satellite map for illustration and annotation, not used by the trajectory predictor; (b) the semantic map after annotation, used by the trajectory predictor; (c) the drone image used to illustrate bird's eye view perception results from the subject vehicle, not used by the trajectory predictor; (d) the vehicle relationship graph, used by the trajectory predictor.



vehicle, we crop the semantic map centered at the subject vehicle and form the graph also centered with the subject vehicle. If there are more than 25 vehicles on the scene, we select the nearest 25 vehicles to form the graph. More details about the graph are explained in the original Trajectron++ paper [21].

As a driver behaves abnormally under a DZ, we can compute the deviation of the driver's trajectory and the model's predicted trajectory:

$$\Delta Path = \sum_{i=1}^n \|y_{pred} - y_{true}\|_2 \quad (4)$$

However, a dilemma cannot be calculated just by thresholding $\Delta Path$. The driver would have made an abrupt lane change, slowed down, or accelerated on purpose, which would deviate from the model's predicted trajectory. An expert classifier is required to classify whether $\Delta Path$ belongs to a DZ event or not. In our detector, we use a shallow neural network that takes as input the path deviations for the next four timesteps ($\Delta 1, \Delta 2, \Delta 3, \Delta 4$), as detailed in Table 1. Other information can also be combined in future work.

Finally, the abnormal driving prediction is made by thresholding z to be 0.5 after training.

TABLE 1 Network architecture for dilemma detection

Layer	Input	Output	Activation	Size
input	$\Delta 1, \dots, \Delta 4$	$\mathbf{h}$	none	4×32
hidden	$\mathbf{h}$	$\mathbf{g}$	sigmoid	32×1
output	$\mathbf{g}$	z	sigmoid	1×1

Dilemma Forecasting and Decision Making

Our forecasting method leverages the state graph that was originally employed in the ST++ trajectory prediction algorithm. This state graph serves as a foundational framework for modeling vehicle movements and interactions within a roundabout. By combining the spatial awareness of dilemma zones with future trajectory forecasts, the model can provide effective dilemma forecasts for approaching vehicles.

Graph Neural Network Preliminaries

To effectively model the non-Euclidean structure of traffic, where the number of vehicles varies and their interactions are unstructured, we model the roundabout scene as a graph $G = (V, E)$. Here, V represents the set of nodes where each node corresponds to an individual vehicle on the scene. The edges E (represented by the adjacency matrix A) denote the spatial interactions or social connections between these vehicles.

A Graph Neural Network (GNN) is employed to process this structure. Unlike standard neural networks that expect fixed-size inputs, GNNs utilize message passing operations to aggregate information from neighbouring nodes (interacting vehicles). This allows the model to update each vehicle's representation based on its surrounding traffic context, ensuring that the final dilemma prediction accounts from both the ego-vehicle's dynamics and the influence of nearby agents.

Network Architecture and Encoding

The architecture for our node classifier is detailed in Table 2. The encoding process functions as follows:

- **Node Features (V, D):** The input feature matrix, where V is the number of vehicles in the current

frame, and D is the feature dimensions (10 dimensions). These features explicitly encode the vehicle's kinematic state (position, velocity, acceleration) and its motion history.

- **Adjacency Matrix (V, V):** A weighted matrix representing the connectivity between vehicles. If two vehicles are close enough to interact, an edge exists between them, allowing the GCN layers to incorporate their features.
- **Graph Convolution (GCN 1 & 2):** These layers apply the graph convolution operation with ReLU activation. They transform the initial node embeddings (size $E = 128$) by aggregating features from connected neighbours, effectively capturing the "social" context of the roundabout.

Prediction Heads

Following the shared feature extraction, the network branches into three separate output heads to assess the likelihood of specific events:

1. **P_{Dilemma} :** the probability that the vehicle node will enter a dilemma zone.
2. **P_{Pass} :** the probability that the vehicle node can pass through the intersection by accelerating.
3. **P_{Causal} :** the probability the vehicle node's behavior causes a dilemma.

FIGURE 6 Dilemma forecasting with a graph neural network.

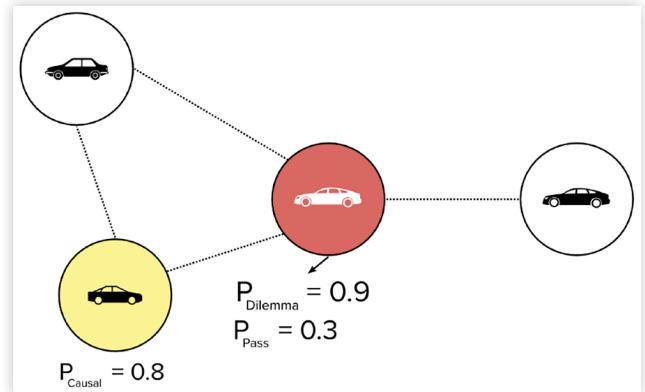


TABLE 2 Dilemma forecasting GNN model architecture

Layer	Input	Output	Details
Node Features	(V, D)	(V, D)	Input graph nodes
Adjacent Matrix	(V, V)	(V, V)	Input graph edges
Graph Embedding	(V, D)	(V, E)	Input to embedding
GCN 1	(V, E)	(V, E)	Graph convolution layer with ReLU
GCN 2	(V, E)	(V, E)	Graph convolution layer with ReLU
Shared Features	(V, E)	(V, E)	Pooled graph features
Dilemma Prediction	(V, E)	($V, 1$)	FC layer with sigmoid activation
Passing Prediction	(V, E)	($V, 1$)	FC layer with sigmoid activation
Causing Prediction	(V, E)	($V, 1$)	FC layer with sigmoid activation

To train our dilemma event forecasting model, we harness the wealth of data generated by ST++. The training process involves feeding the trajectories with the information of them falling into a DZ into our forecasting model and fine-tuning the model's parameters. The loss function optimizes all three prediction heads jointly. Backpropagation updates the shared parameters to extract relevant features, and the individual heads to predict the corresponding probabilities. This iterative process refines the model's ability to identify potential

DZ events based on real-world trajectories and empirical data.

During inference, the model takes in a new vehicle node's attributes and outputs the three predicted probabilities. These probabilities are used to determine if the node is likely to encounter a dilemma, is unlikely to pass through, or could potentially cause a conflict. Based on predefined thresholds, the system can forecast impending dilemmas for vehicles approaching the intersection. Here, we choose the threshold to be 0.5, since the sigmoid prediction is always a probability from 0 to 1. If the predicted P_{Dilemma} is above 0.5, it means that our model concludes this vehicle is more likely in a dilemma zone than free from a dilemma zone. Multi-task learning with a shared feature extractor and three separate output heads enables efficient dilemma forecasting using graph node classification. The three predicted probabilities provide comprehensive situational awareness for the subject vehicle.

Experimental Evaluation

To validate the proposed methods for dilemma zone determination and abnormal driving event detection, we conduct a series of experiments using real-world vehicle trajectory data. This section describes the experimental setup, implementation details, and ablation studies performed to assess model performance and the impact of dilemma zone features. The experiments evaluate both the accuracy of trajectory prediction and the effectiveness of dilemma zone detection under various driving scenarios at roundabouts.

Experiment Setup

Dataset: We evaluated our methods using the publicly available CAROM Air dataset curated by Lu et al. [13]. This dataset is a collection of videos of vehicle trajectories in the United States recorded by a drone. There are diverse sites covered in the dataset, such as roundabouts, intersections, local road segments, and highways. The dataset has a variety of features, including the local position coordinates of vehicles, their velocity and acceleration, and semantic maps of the intersections. Each site has approximately 60 minutes of footage, and there are approximately 4-5 vehicles in each frame. We use the vehicle trajectory data of ten roundabout sites in the dataset.

To calculate dilemma zones on the road surface, we use the speed limits suggested by the U.S. Department of Transportation [25]. The speed is constrained between 15-25 Mph for any oncoming vehicle, irrespective of the roundabout and regardless of the posted speed limits on approaches.

We calculate the dilemma zone by using [equations \(1\) and \(2\)](#). We plug in the speed limit of the intersection as the velocity of the vehicle and the safe deceleration or braking rate as a_{dec} [17].

Baselines: To compare the modified trajectory prediction model, we use the original Trajectron++ as the baseline, which is trained without DZ features on the same dataset.

Metrics: For dilemma zone event detection, two sets of metrics are used as follows.

1. False Positive Rate (FPR) is used as the main metric to evaluate the dilemma zones detected by the system. This is used as the primary metric for DZ classification, as the model would face a high number of non-dilemma scenarios, as dilemma zone events are rare. Additionally, the F1 score and recall are used to show the overall performance of the detector.
2. Intersection over Union (IoU) measures the overlap between the detected dilemma zones and the ground truth dilemma zones.

For trajectory forecasting, two metrics are used as follows.

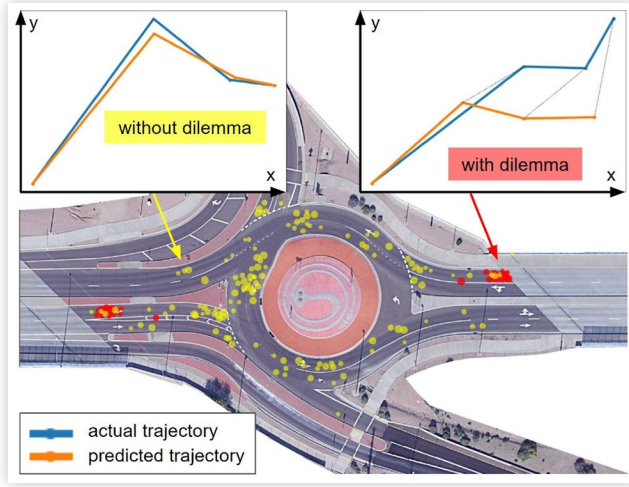
1. Average Displacement Error (ADE), which is the average of L2 norms between the i th position of the ground truth and the prediction horizon. ADE measures the overall trajectory accuracy across time, capturing how well the predicted path follows the ground truth throughout the entire horizon rather than focusing on a single instant.
2. Final Displacement Error (FDE), which is the L2 norm between the predicted final position and the ground truth final position at the prediction horizon. FDE emphasizes endpoint accuracy, reflecting the model's ability to correctly anticipate the agent's final intent or destination.

Implementation Details

We implement the system with Python and PyTorch [26]. Using the dataset, we demonstrate the trajectory prediction model and detection of dilemma zones on Roundabout-A2, as the number of dilemma events is considerable. 80% of the video is randomly sampled for training, whereas the test and validation sets are each 10%. Before training, the dilemma zones are calculated from the training video by exhaustively checking each vehicle pair using Algorithm 1 in [Figure 4](#). The ST++ is trained for 100 epochs with a batch size of 64. The DZ classification network is trained for 100 epochs with a batch size of 128. All the models are trained on a single Nvidia GeForce GTX 1080 GPU. The hyperparameters of the model are set to $H = 3$, Time of entry $T = 4$, $T_{\text{max}} = 1.5$ s, $D_t = 10$ m, $\alpha = 0.5$, $\gamma = 0.2$, where H and T are the number of timesteps in the input trajectory and the number of timesteps in the prediction window, respectively. We threshold P_{Pass} and P_{Dilemma} at 0.5 for dilemma forecasting.

To evaluate the effectiveness of our proposed method, we observe the trajectory deviation of a vehicle in a DZ versus that of a non-DZ, as shown in [Figure 7](#).

FIGURE 7 Abnormal driving events detected using trajectory deviation from the prediction with dilemma (red) and without dilemma (yellow).



The trajectory deviation is defined as the ratio of the L2 norm of the difference between the predicted trajectory and the ground truth trajectory to the L2 norm of the ground truth trajectory. All the path deviations that had a deviation larger than 0.8 (presence of a large deviation) at one of the predicted timesteps were collected. But, due to the presence of only a small portion of the deviations that belonged to the DZ (20% of Non-DZ), we do account for an equal portion of the largest deviations in the non-dilemma zone. This was used as the dataset to train the shallow network listed in Table 1. The results are shown in Table 3.

Ablation Study

Effects of Dilemma Zone Features: We evaluate the effects of the DZ features in the process of detecting dilemma zones. We use the vanilla version of Trajectron++ without DZ features to observe the effects of the DZ features. The system's performance without DZ features

is compared with DZ features based on the metrics defined previously. The results are shown in Table 4. The results show that the addition of DZ features improves trajectory prediction by an effective margin. The ADE and FDE are reduced by 5.15% and 7.45% respectively, with the addition of the DZ features.

By observing the receiver operating characteristic (ROC) curve shown in Figure 8, it can be seen that the classifier works better with our model that has DZs. The FPR is reduced greatly by 30.45% with the addition of DZ features. These results show that the DZ features play a significant role in detecting dilemma zones.

Path Deviation under Dilemma: Figure 9 shows the average trajectory deviation of vehicles inside a DZ and those that are not. It can be observed that the trajectory deviation of a vehicle inside a dilemma zone is higher than that of a non-dilemma zone. Compare the average trajectory deviation of vehicles that are in a dilemma zone to those that are not.

FIGURE 8 Trajectory Forecasting Model Comparison: The proposed superpowered trajectron++ outperforms the baseline trajectron++ model as depicted by the ROC curves. The deviations estimated from our method exhibit clear distinctions between dilemma and non-dilemma scenarios.

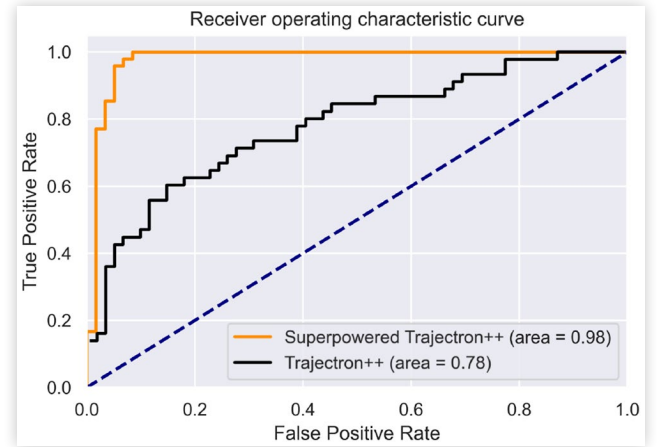


TABLE 3 Evaluation results of the dilemma detection and forecasting methods.

task	dataset	F1 score	recall	FPR	IoU
DZ detection	training	0.93	0.92	0.14	0.92
	validation	0.92	0.93	0.06	0.89
	testing	0.91	0.95	0.10	0.87
DZ forecasting	training	0.91	0.94	0.09	-
	validation	0.88	0.90	0.14	-
	testing	0.83	0.81	0.17	-

TABLE 4 Evaluation results of trajectory prediction models.

method	ADE (m)	FDE (m)				FPR
		@0.5sec	@1 sec	@1.5 sec	@2 sec	
Trajectron++	2.252	0.144	0.603	1.487	2.535	0.157
Ours	2.136	0.183	0.604	1.401	2.346	0.109

FIGURE 9 Dilemma vs. Non-Dilemma: A comparison of the average trajectory deviation in a yield zone.

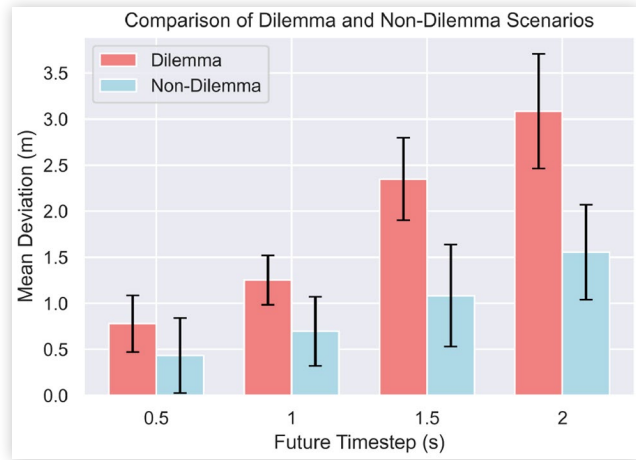


TABLE 5 Ablation study results of maneuvering prediction.

prediction	% of collision-free cases given the acceleration			
	2 m/s ²	4 m/s ²	-2 m/s ²	-4 m/s ²
$P_{Pass} > 0.5$	83	94	3	21
$P_{Pass} \leq 0.5$	5	6	91	79

Maneuvering: The maneuver was implemented only on the ego vehicle. The acceleration of the ego was increased or decreased while the rest of the vehicles continued to follow the trajectory predicted by ST++. The simulation is run on 200 randomly sampled cases of the dilemma, with 100 scenarios in which the vehicle has to accelerate and clear the intersection, and vice versa. According to the US Federal Highway Authority [27], the maximum limit of safe acceleration is 4 m/s², which is considered the upper limit in this maneuvering experiment.

The results are shown in Table 5. The key observations are that vehicles need to accelerate rapidly when there is an opportunity to pass, and when the option to stop is present, deceleration should be gradual.

Conclusions and Future Work

This study introduces a novel perception-based system aimed at detecting and forecasting dilemma zones at roundabouts through trajectory prediction. By analyzing trajectory deviations in dilemma zones compared to non-dilemma zones, the system can anticipate when a vehicle is in a dilemma with a low FPR of 10%. However, certain limitations are acknowledged, including the system's reliance on data quality, its specific applicability to roundabouts, and potential computational challenges in real-time performance. Despite these constraints, this work lays a foundation for future advancements in traffic safety and management. We aim to extend the framework

beyond roundabouts to various scenarios like unsignalized T-bone intersections and ramp-merging, enhancing real-time forecasting capabilities for intelligent transportation systems. Additionally, implement forecasting and maneuver techniques in real-world controlled scenarios to proactively address potential hazards and optimize vehicle responses and overall safety. Our team is currently working with the Arizona Institute of Automated Mobility and the City of Phoenix to seek real-world experiments of the presented system in the Phoenix metropolitan area. We hope this paper can collectively enhance our understanding of DZ dynamics and pave the way for more robust and proactive solutions in autonomous vehicle safety.

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Appendix A: A List of Roundabouts in the CAROM Air Dataset

